CITYINSIGHT: WEB AND MOBILE-BASED PERSONAL SAFETY COMPANION

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Index Terms
Crime Monitoring , Flood Detection , Emergency Alerts , Real-Time Safety Navigation , Urban Safety Solutions , CityInsight Application , Hazard Awareness , Sumi Usability

Abstract - Crime and flooding are significant concerns in Urdaneta City, affecting public safety and navigation. This study presents **CityInsight**, a comprehensive web and mobile-based application designed to provide real-time alerts on hazards and emergency services. The app integrates features like crime policy updates and flood-prone area notifications to help users make informed decisions, avoid dangerous areas, and feel safer in their daily activities. Developed using SUMI methods to prioritize user needs, the system ensures efficiency and reliability through usability testing and iterative improvements. Data collection guided the inclusion of essential features tailored to Urdaneta City's unique challenges. CityInsight offers a user-friendly interface, real-time updates, and tools that enhance safety and navigation during emergencies. Testing confirmed its effectiveness in simplifying decision-making and bolstering confidence in personal safety. The study highlights the potential of technology to address urban safety issues and provides a scalable framework for developing similar solutions in other regions, aiming to foster safer and more secure communities.

I. INTRODUCTION

Safety and resilience are critical aspects of urban living, especially in cities like Urdaneta City, which faces the dual challenges of crime and flooding. The interplay between human activities and natural events often results in hazards that significantly impact communities, households, and organizations. Crime, often unpredictable, can destabilize communities already burdened by other shocks such as flooding, thereby hindering sustainable development (Kwanga, Shabu, & Adaaku, 2017). Urban safety is further compromised by Urdaneta City's geographical susceptibility to flooding, exacerbated by rapid urbanization and inadequate early warning systems.

Efforts to address these challenges are evident in tools like the Philippine National Police’s Crime Incident Reporting System (CIRS) and Crime Information Reporting and Analysis System (CIRAS). These systems provide real-time crime data and mapping, aiding law enforcement in identifying patterns and deploying resources effectively. Similarly, Geographic Information Systems (GIS) have been integral to flood management, offering precise flood hazard maps and supporting early warning systems. Yet, localized solutions are needed to address the specific safety concerns of Urdaneta City.

To fill this gap, **CityInsight** emerges as a web and mobile application designed to provide real-time updates on crime and flood-prone areas. By integrating community-sourced data and advanced technologies, the app aims to empower individuals with actionable insights to navigate their surroundings safely and confidently. CityInsight serves as a companion in mitigating uncertainties related to urban hazards, demonstrating the potential of technology in addressing complex safety issues and fostering resilient urban communities.

A. Statement of the Problem

One of the critical challenges in urban environments is ensuring public safety and efficient navigation, particularly in areas prone to crime and natural hazards. Citizens often lack timely information about potential risks, leaving them vulnerable to safety threats and logistical difficulties. Furthermore, traditional systems do not integrate real-time updates, limiting their effectiveness in providing actionable safety insights and navigation support. The research aims to design and develop CityInsight: Web and Mobile-based Personal Companion to address these issues. CityInsight will deliver real-time safety updates, such as alerts about hazards and emergency services, empowering users to make informed decisions, avoid risky areas, and enhance their confidence while exploring new locations.

Objectives this study was designed to solve the following problems: 1. How can we identify and implement effective policies and procedures to respond to various crime incidents in Urdaneta City? 2. What technologies and strategies can be used to gather and present real-time data about flood-prone areas, ensuring timely and reliable safety updates? 3. What features and functionalities are necessary for CityInsight to enhance user safety and improve navigation experiences effectively?

II. RELATED LITERATURE

The Philippine National Police (PNP) has implemented a systematic approach to crime incident response, utilizing protocols that prioritize timely reporting and coordinated action. This includes the deployment of specialized units like SWAT teams and anti-crime task forces. The PNP’s centralized Crime Incident Reporting System (CIRS) enables the mapping and analysis of criminal activities to identify trends and improve resource allocation. Additionally, the integration of the Crime Information Reporting and Analysis System (CIRAS) has enhanced the visualization of crime hotspots, supporting strategic interventions to reduce crime rates.

Parallel to crime response efforts, the National Disaster Risk Reduction and Management Council (NDRRMC) employs detailed flood risk assessments, early warning systems, and collaborative response plans to mitigate flood-related risks. By mapping flood-prone areas and coordinating with local government units (LGUs), the NDRRMC aims to enhance community preparedness and resilience. These protocols highlight the importance of real-time data, resource coordination, and community involvement in addressing public safety and disaster management challenges.

Urban environments present numerous challenges when it comes to personal safety and disaster preparedness. Many residents lack access to timely, reliable information about crime hotspots or areas vulnerable to natural disasters such as floods, leaving them at a greater risk of harm. Research shows that this absence of situational awareness leads to poor decision-making, such as unknowingly entering high-crime areas or traveling through flood-prone streets during adverse weather conditions, which can jeopardize lives and property. Additionally, limited access to real-time updates, fragmented safety protocols, and unclear emergency response procedures compound the difficulties faced by individuals during critical moments. Communities often rely on outdated methods of disseminating safety information, which can result in delays and confusion during emergencies.

CityInsight addresses these challenges by harnessing the power of technology to deliver an all-in-one safety companion platform. Through its integration of GIS mapping, real-time data analytics, and community-driven reporting, the app empowers users with up-to-date information about their surroundings. For example, CityInsight provides dynamic alerts about nearby safety risks, such as crime incidents or flooding, and offers alternative routes or guidance to avoid these hazards. By incorporating predictive analytics, the platform also identifies emerging risks, enabling users to make proactive decisions to protect themselves and their families.

The app's features go beyond alerts by promoting community engagement and awareness. Residents can report safety concerns directly within the app, contributing to a crowd-sourced database that enhances situational awareness for everyone. CityInsight also connects users to relevant emergency services, providing quick access to assistance when it is needed most. Moreover, educational resources within the app offer valuable information on preparedness strategies, such as how to respond to natural disasters or mitigate risks in high-crime areas, fostering a culture of safety and resilience.

By leveraging technology and community collaboration management Its innovative approach not only provides immediate benefits to users but also contributes to long-term improvements in public safety by reducing risks, enhancing preparedness, and fostering trust in technology-driven solutions. Through tools like CityInsight, cities can become safer, more informed, and better equipped to handle the challenges of modern urban living.

III.METHODOLOGY

A. Research Design

The research study will employ developmental and descriptive research methodologies to design, develop, test, and evaluate CityInsight, a web-mobile application designed to provide insights into crime rates and safety through crowdsourced information. The integration of these methodologies will be crucial in achieving significant findings during the study.

The research process began with a planning phase, where the objectives were clearly defined, and the appropriate research tools were developed. This included the creation of a structured survey questionnaire instrument, which was meticulously designed to gather detailed information from respondents. To gain deeper insights, the study also incorporated interviews with key stakeholders and residents to capture qualitative data on safety concerns and perceptions within specific barangays in Urdaneta City. Additionally, the Likert Scale methodology was employed to quantitatively assess the level of safety concerns, providing a robust analysis of the issues at hand.

B. Research Setting and Participants

The study will be conducted around Urdaneta City, Pangasinan. To ensure a comprehensive understanding of the functional requirements of the proposed system, we will use purposive sampling to select 30 participants from Urdaneta City. This method allows us to intentionally include key stakeholder groups—residents, students, police officers, and NDRRMC officers—who possess specific knowledge and perspectives relevant to the system. By focusing on these targeted groups, we ensure that all critical viewpoints are considered, providing a nuanced and informed basis for the system's design. The selected sample size of 30 participants allows for in-depth qualitative analysis. By focusing on a smaller, targeted group, we can gather rich, detailed insights that are crucial for understanding the specific functional requirements of the proposed system.

Table 1. *Distribution of Respondents*

Respondent	**Objective 1**	**Objective 2**	**Objective 4**
Police In-Charge	**1**	**5**	**5**
NDRRMC Respondents	1	5	5
Urdaneta Residents	0	10	10
Urdaneta City Students	0	10	10
Total	2	30	30

Data were collected using three instruments for the study. The first was online surveys, which collected demographic information. Using structured interview as a data collection method, the researchers focus on the policies and procedures for accurately disseminating crime incident data. It will also aim to gather detailed information about flood- prone areas in Urdaneta City. The researchers will ensure that all collected information is precise and reliable as this structured approach will provide a solid factual foundation for developing the system, ensuring that it is based on comprehensive and accurate data.

Finally, quantitative data on the user experience of the CityInsight prototype was obtained with the SUMI (Software Usability Measurement Inventory).

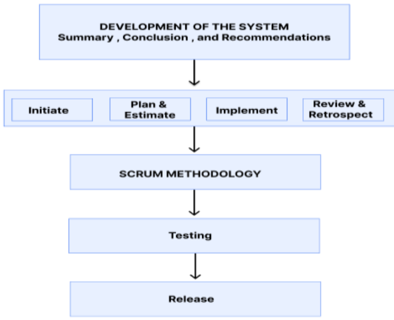

Figure 1. *Flowchart Diagram for Proposed Implementation Plan*

The CityInsight system was built with SUMI Methodology, which allows for an incremental and adaptive process. This process consisted of: Initiation: in this phase the researchers defined project vision, goals and deliverables. Working together with local authorities, we learned more about Urdaneta City Pangasinan Police Station. The team categorized high-level features such as available slot monitoring in Crime Monitoring, Flood Detection, Emergency Alerts Real-Time Safety Navigation, Urban Safety Solutions, Hazard.

Planning and Estimation, During the implementation phase, the application's researchers will employ Flutter to build the system's mobile section, with the goal of providing smooth and user-friendly experiences. In terms of online applications, they want to leverage PHP to provide robust and dynamic functionality on the website. This combination is expected to create a versatile and effective system that caters to diverse users across devices and platforms.

The *implementation* phase is focused on executing the planned tasks to achieve the objectives of the project. **CityInsight** integrates data collection from various sources, including local authorities and publicly available databases, to provide real-time safety updates. Advanced geolocation and mapping technologies were used to deliver precise hazard alerts, such as crime-prone zones and flood-prone areas.

During the *review and retrospective* phase, **CityInsight** was tested by potential end-users, including residents, local authorities, and students in Urdaneta City. Feedback on usability, such as the clarity of safety alerts and the responsiveness of the interface, was gathered and analyzed.

The system's features were iterated based on this feedback to improve user experience, such as refining real-time updates on crime and flood-prone areas and enhancing map navigation functionalities. Lessons learned during regular retrospectives were documented to ensure continuous improvement and to increase the system's reliability and acceptance by the users

Release: The researchers will release the program's initial release, which will mark a significant milestone in the development process. This step involves launching the system and making it available to the intended user base. Prior to launch, the application will undergo significant planning, configuration, testing, and validation to ensure its readiness. With all of this protection in place, the release phase marks the beginning of users' interactions with the software, resulting in actual event usage and impact. This phase not only signals the conclusion of development, but it also provides for user participation and data collecting. As users experience the app's features, their feedback is critical for future enhancements and modifications.

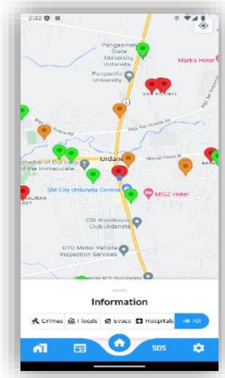

Figure 2: *Home Screen of the Mobile Application for End Users*

The Home Screen includes the real-time location, navigate evacuation centers and hospital using polylines and the map that shows the information about crimes, floods, and evacuation center.

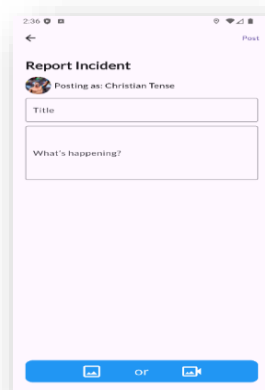

Figure 3: *Report Incident of the Mobile Application for*

End Users

The Report Incident include the report of crime or flood incident the end user can upload images and videos.

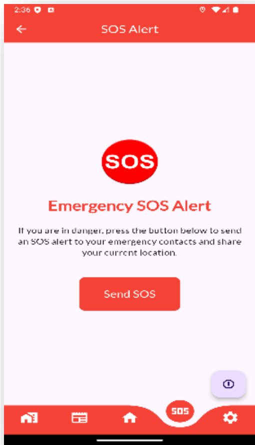

Figure 4: *One Tap Alert of the Mobile Application for End Users*

The One Tap Alert include the button allows users to quickly report emergencies or request assistance with a single tap, streamlining the process for immediate response and support.

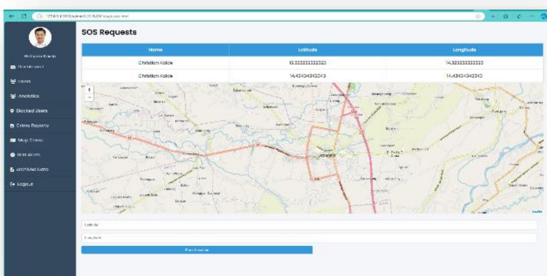

Figure 5: *SOS Request of the Users to the Police*

The SOS request allows the police to see quickly the report emergencies or request assistance and streamlining the process for immediate response and support.

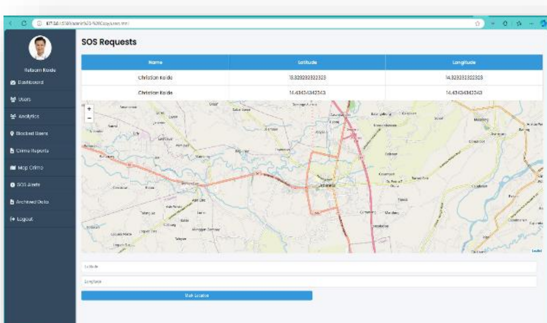

Figure 6: *SOS Request of the Users to the CDRRMC*

The SOS request allows the NDRRMC to see quickly the report emergencies or request assistance and streamlining the process for immediate response and support.

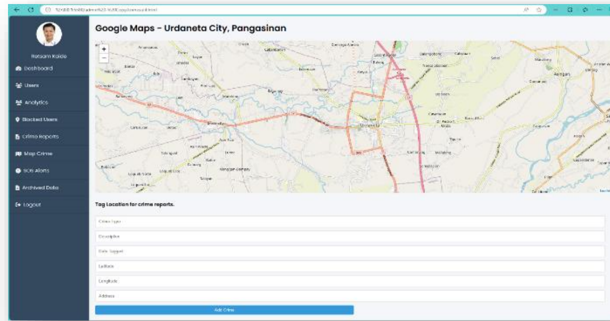

Figure 7: *Adding the Flood Incidents of CDRRMC*

The adding flood incidents include the flood type, description, date tagged, latitude and longitude, and the address.

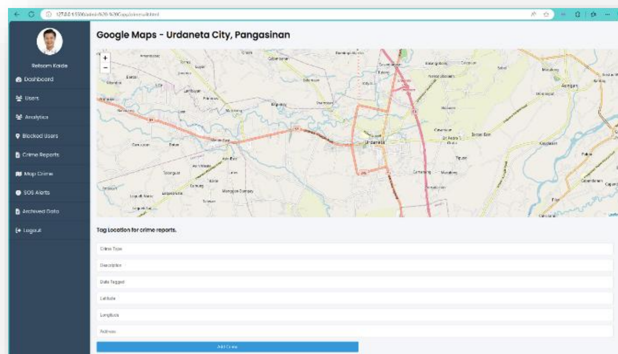

Figure 8: *Adding the Crime Incidents of the Police*

The adding crimes incidents include the crime type, description, date tagged, latitude and longitude, and the address.

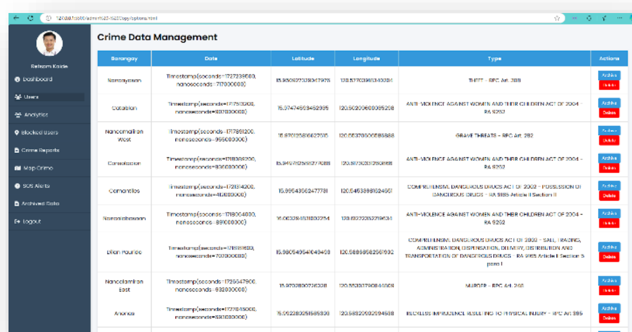

Figure 8: *Adding the Crime Incidents of the Police*

The crime data management allows the police to archive and delete the reports.

IV.RESULTS

This study examines the challenges and inefficiencies in addressing crime incidents and flood hazards in Urdaneta City and explores how CityInsight, a web and mobile-based application, can effectively mitigate these issues.

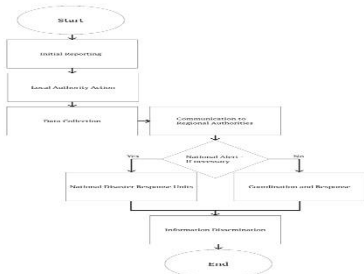

Figure 9. *Flow Chart for the Traditional Process in Terms of Responding to Natural Disasters*

Figure 4 illustrates the traditional disaster response process of the CDRRMC, which follows a linear progression. It begins with issuing disaster alerts, followed by impact assessment to evaluate damages and risks. Based on the assessment, resources are mobilized, and response teams are deployed for rescue and relief operations. The process concludes with recovery and rehabilitation, focusing on rebuilding and restoring normalcy.

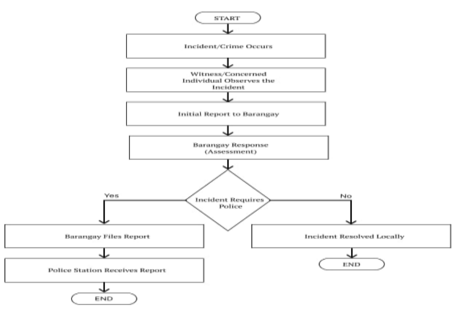

Figure 10. *Flow Chart for the Traditional Process in Terms of Incident or Crime Reporting*

Figure 6 depicts the traditional process for reporting incidents or crimes in Urdaneta City, Pangasinan. It starts with the occurrence of an incident, followed by a witness reporting it to the Barangay. The Barangay assesses the situation to determine if police involvement is required. If necessary, a report is filed with the local police; otherwise, the incident is resolved locally within the Barangay. The flowchart emphasizes key decision points, such as the need for police intervention.

The functional requirements for the CityInsight system were analyzed based on feedback from 30 respondents. Features prioritized for development include location tagging (100%), crime or disaster mapping (83.33%), and crime heat maps (70%). Additional features such as real-time location tracking (66.67%), incident reporting (66.67%), analytics (60%), and one-click emergency services (60%) were also highly requested. Features requested by fewer than 50% of respondents will be considered optional. This prioritization ensures that the development focuses on the most critical features identified by users, aligning with their needs and expectations for the system.

Using the SCRUM method, CityInsight was developed as a mobile and web-based application. The mobile app provides users with real-time updates, location tagging, crime or disaster mapping, and one-click emergency reporting, enabling better decision-making during emergencies. Meanwhile, the web platform offers administrators tools for monitoring incidents, analyzing data, and managing alerts effectively. This iterative, user-centered approach ensured a scalable and efficient system tailored to the unique safety and navigation needs of Urdaneta City.

Table 2. Overall Result of Usability Test

Dimension	Mean Score	Descriptive Equivalent	Descriptive Interpretation
Efficiency	3.00	Moderately Agree	Usable
Affect	2.88	Moderately Agree	Usable
Control	3.34	Agree	Usable
Helpfulness	3.15	Agree	Usable
Weighted Mean	**3.01**	**Moderately Agree**	**Usable**

Mean Score Range	Descriptive Equivalent	Descriptive Interpretation
4.30 – 5.00	Strongly Agree	Usable
3.50 – 4.29	Agree	Usable
2.50 – 3.49	Moderately Agree	Usable
1.70 – 2.49	Disagree	Not Usable
1.00 – 1.69	Strongly Disagree	Not Usable

Table 3. *Likert Scale for Usability Testing*

Table 2*:* Usability testing using the System Usability Scale (SUS) for CityInsight resulted in a high score of 3.09, classified as "Moderately Agree" and "Usable." Users were impressed by the system’s intuitive layout, smooth navigation, and key features such as real-time location updates and crime mapping. Minor issues, such as occasional delays during peak usage, were addressed through system optimization

The results validate CityInsight as an effective and user-friendly solution for improving urban safety. This aligns with existing literature on smart city applications, including Smith et al. (2023), which emphasized the importance of real-time alerts in enhancing public safety and decision-making [11]. Additionally, Adams et al. (2019) demonstrated how real-time mapping and data-driven insights could improve emergency response effectiveness [12]. CityInsight effectively addressed safety concerns, provided better navigation, and offered valuable tools for administrators to monitor and manage incidents.

The high usability score (3.09) suggests a promising future for urban safety solutions. CityInsight enhances the overall safety experience for residents by helping them navigate potential risks, receive real-time updates, and make informed decisions. The system's user-friendly data management tools are intuitive and easy to use, ensuring quick and efficient analysis for emergency response teams and administrators.

V.CONCLUSION

Urban safety and navigation are persistent challenges, and Urdaneta City is no exception. The need for real-time updates on crime incidents, flood-prone areas, and emergency alerts has become increasingly important for residents and visitors alike. However, recent developments in smart city solutions have led to the creation of CityInsight, which effectively addresses these issues. CityInsight helps in improving public safety by providing real-time crime and disaster mapping, along with features like location tagging and incident reporting, offering residents the information they need to navigate their daily activities safely.

The development of CityInsight was guided by agile software development practices, using the SCRUM methodology. This iterative approach allowed the team to break down the development process into manageable cycles: planning, development, and review. Through continuous feedback and improvement, the system was designed to meet the needs of users while ensuring that the final product was efficient, scalable, and user-friendly.

In addition to the successful implementation of the CityInsight system, extensive usability testing revealed very high satisfaction ratings. The results indicate that the system is highly usable, with users noting its intuitive design and real-time functionalities. CityInsight has proven to be an effective solution in enhancing safety and navigation for Urdaneta City's residents and is positioned as a model for how municipalities can use technology to address urban safety issues.

In addition to the successful implementation of the CityInsight system, extensive usability testing revealed very high satisfaction ratings. The results indicate that the system is highly usable, with users noting its intuitive design and real-time functionalities. CityInsight has proven to be an effective solution in enhancing safety and navigation for Urdaneta City's residents and is positioned as a model for how municipalities can use technology to address urban safety issues.

In conclusion, CityInsight not only solves the pressing challenges of urban safety and navigation but also showcases the power of SCRUM methodology in software development. Through its user-centric design and functionalities, CityInsight is revolutionizing how urban communities approach safety and emergency response, setting a new standard for smart city solutions.

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